



DSD Project Week 4 Report

Group 5 Project: Frequency Difference Estimator

Role	Name	Student ID
A	Alireza Shakeri	401101881
B	Fatemeh Amini	402101299
C	Arshia Noroozi	402170562
D	Mahdi Emami	400100702

Digital System Design with Verilog (Spring 1405)

Instructor: F. Baharvand

Sharif University of Technology

Electrical Engineering Department

Week 4 – Final Report

Module Report – Frequency Difference Estimator

Group ID: G5

Members: A: Alireza Shakeri, B: Fatemeh Amini, C: Arshia Noroozi, D: Mahdi Emami

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1 Executive Summary

- **Module implemented:** Frequency Difference Estimator (Frequency Model)
- **DRS version:** v1.0
- **Status:** Complete. The final RTL, Version C (`freq_diff_estimator_vC.v`), has passed functional verification, logic synthesis, and gate-level simulation, all completed during Week 3. Week 4 activities consist of system integration and final submission packaging, as outlined in Section 8.

The Frequency Difference Estimator computes the Main RO total relative frequency error ϵ_{RO} as a quadratic function of the temperature deviation from 25°C, combined with an aging contribution supplied by the rRO Aging Engine. The result is represented in signed Q1.23 format and drives the downstream Sigma-Delta Modulator.

The Week 3 objectives – running logic synthesis, analyzing standard-cell area and gate composition, checking the design against the available clock constraint, identifying area-dominant arithmetic structures, investigating alternative RTL implementations, selecting the smallest implementation that preserves functionality and timing, and running post-synthesis gate-level simulation – were all completed, as detailed in Section 4 and Section 5 of this report.

During Week 3, four candidate RTL implementations of the quadratic temperature term were synthesized and compared: Version B (an area-oriented low-resolution approximation), Version A (an original-resolution reciprocal-constant multiplier), Version A2 (a higher-resolution reciprocal-constant multiplier), and Version C (an algebraically simplified exact-fraction implementation). Version C was selected as the final RTL because it combines exact numerical accuracy with the smallest area and power among the accurate implementations. It was verified against the official course testbench across 1,000,097 vectors with zero recorded failures and zero LSB difference from the golden reference model, synthesized to 69,900.971 μm^2 using Cadence Genus with the TSMC 180 nm typical library, and confirmed by zero-delay and SDF-annotated gate-level simulation with zero functional mismatches.

2 RTL Implementation



Figure 1: Initial block diagram of the Frequency Difference Estimator module.

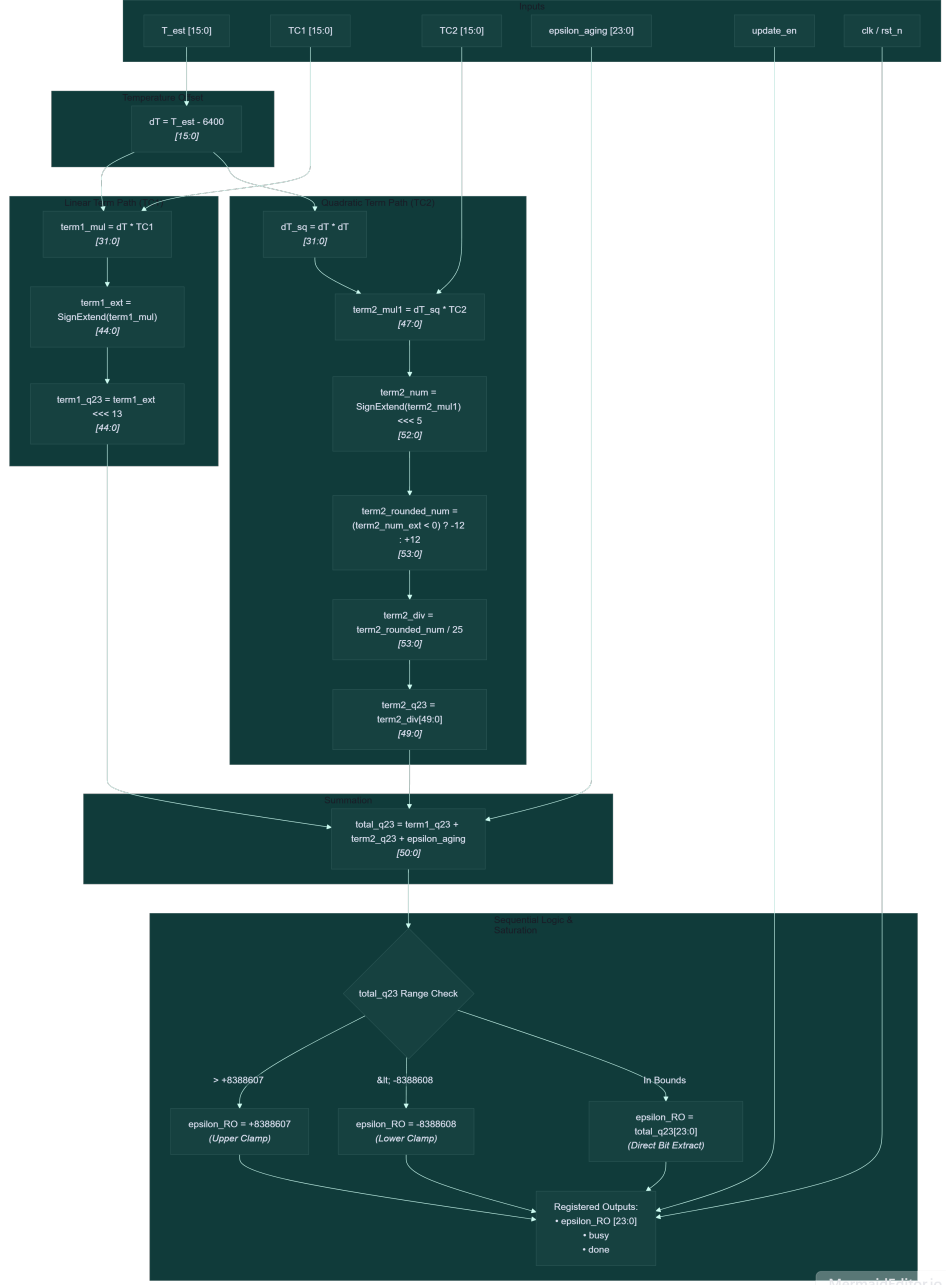


Figure 2: Final implemented datapath (Version C) of the Frequency Difference Estimator module.

- **Total lines of code:** Approximately 84 lines of RTL (`freq_diff_estimator_vC.v`) and approximately 347 lines of testbench code (`tb_freq_diff_estimator.v`).
- **Finite state machines:** None. The module is a single-cycle combinational datapath followed by one synchronous output register; no multi-cycle or FSM-based architecture is present.
- **Key design decisions:**
 - The temperature deviation $dT = T_{est} - 25^{\circ}\text{C}$ is computed directly in Q8.8 format by subtracting the constant $25 \times 2^8 = 6400$ from the raw T_{est} input.

- The linear term is computed as an exact scaling, $\text{term1_q23} = (dT \times TC1) \ll 13$, matching the DRS golden model with no rounding required.
- The quadratic term’s fixed-point scaling was derived algebraically from the DRS quadratic-term equation, rather than approximated with a reciprocal constant:

$$\epsilon_{TC2,Q23} = TC2 \times dT^2 \times \frac{2^{23}}{100 \times 2^{16}} \quad (1)$$

$$= TC2 \times dT^2 \times \frac{2^7}{100} \quad (2)$$

$$= TC2 \times dT^2 \times \frac{128}{100} \quad (3)$$

$$= TC2 \times dT^2 \times \frac{32}{25}. \quad (4)$$

This exact factor is realized in the RTL by multiplying the intermediate product $dT^2 \times TC2$ by 32 (an exact 5-bit left shift), rounding to the nearest integer, and dividing by 25 using Verilog’s true integer division operator. This exact-fraction approach was adopted after three earlier reciprocal-constant approximations (Versions B, A, and A2) were found to trade off area against numerical accuracy; the full history is documented in Section 7.

- The output ϵ_{RO} is saturated synchronously to the signed 24-bit Q1.23 range $[-2^{23}, 2^{23} - 1]$.
- **busy** and **done** assert together for exactly one clock cycle following an asserted **update_en**, and are driven directly from the registered **update_en** condition since the datapath is purely combinational.
- $TC1$ and $TC2$ are treated as static combinational inputs, loaded from OTP at power-on and held stable during normal operation. ϵ_{aging} is treated as a pre-scaled signed Q1.23 value, summed directly with the temperature terms with no additional scaling.

3 Testbench and Verification

3.1 Golden Reference Model

A self-checking testbench was built around a floating-point golden reference model, implemented as a Verilog function that computes ϵ_{RO} using real-number arithmetic rather than the fixed-point RTL datapath, so that RTL results could be checked against an independent, high-precision implementation of the DRS equation:

```
1 function integer compute_epsilon_RO_golden;
2     input signed [15:0] T_est_q8;
3     input signed [15:0] TC1_in;
4     input signed [15:0] TC2_in;
5     input signed [23:0] eps_aging_q23;
6
7     real T, TC1_real, TC2_real, eps_aging, eps;
8     real eps_q23_real;
9     begin
10         T = T_est_q8 / 256.0;
11         TC1_real = TC1_in * 0.25;
12         TC2_real = TC2_in * 0.01;
13         eps_aging = eps_aging_q23 / 8388608.0;
14
15         eps = (TC1_real * (T - 25.0))
16             + (TC2_real * (T - 25.0) * (T - 25.0))
17             + eps_aging;
18
19         // Q23
20         eps_q23_real = eps * 8388608.0;
21
22         if (eps_q23_real > 8388607.0)
23             eps_q23_real = 8388607.0;
24         else if (eps_q23_real < -8388608.0)
25             eps_q23_real = -8388608.0;
26
27         if (eps_q23_real >= 0)
28             compute_epsilon_RO_golden = $rtoi(eps_q23_real + 0.5);
29         else
30             compute_epsilon_RO_golden = $rtoi(eps_q23_real - 0.5);
31     end
32 endfunction
```

The golden model performs the temperature-error calculation directly in double-precision real arithmetic – converting `T_est`, `TC1`, `TC2`, and `epsilon_aging` out of their fixed-point representations, evaluating the quadratic DRS expression with no intermediate rounding, and re-quantizing and saturating the result to signed Q1.23 only at the final step. This makes it independent of any of the fixed-point scaling choices explored in RTL Versions B, A, A2, and C, so it serves as an unbiased reference for the `check_vector` self-checking task, which applies each input vector to the DUT, checks the `busy/done` handshake, computes `compute_epsilon_RO_golden`, and compares it against the RTL output within the required tolerance.

- **Number of test cases:** 1,000,097 vectors in the official RTL regression, consisting of:
 - 5 directed test vectors;
 - 1 aging-injection check;
 - 81 exhaustive boundary/corner combinations ($3 \times 3 \times 3 \times 3$ over temperature, `TC1`, `TC2`, and aging);

- 2 saturation checks;
- 8 targeted cancellation-case vectors;
- 1,000,000 constrained-random vectors.

For gate-level simulation, the regression depth was reduced to 1,097 vectors (97 directed/boundary/saturation/cancellation vectors plus 1,000 constrained-random vectors), since replaying all 1,000,000 random vectors at gate level was impractical.

- **Coverage achieved:** No percentage-based statement/branch/FSM coverage was available from the verification results.
- **Verification methodology:** Both directed and constrained-random verification were used against the golden model above. Directed testing targeted known, specification-derived behaviors (reset, nominal operation, coefficient sweeps, the handshake protocol, and saturation limits). Constrained-random verification was layered on top specifically to search for rare cancellation-sensitive numerical errors, where the linear and quadratic temperature terms are individually large but nearly cancel.

3.2 Functional Comparison Across RTL Versions

Table 1 summarizes the functional regression results for all four RTL candidates evaluated during Week 3 and 4, each compared against the golden reference model above.

Table 1: Functional comparison of the evaluated RTL versions.

Version	Quadratic scale	$\approx$ Decimal	Max. error	Result
B	$328/2^8$	1.28125	Exceeded 20 ppm	Rejected
A	$171,798,692/2^{27}$	1.280000001192	2,569 LSB	Rejected
A2	$43,980,465,111/2^{35}$	≈ 1.28	3 LSB	Passed (area cost)
C	32/25	1.28 (exact)	0 LSB	Selected

Version A2 passed the full 1,000,097-vector regression with a maximum numerical difference of 3 LSBs (0.357628 ppm). Version C, the final RTL, passed the same regression with zero recorded failures and a maximum numerical difference of 0 LSBs (0 ppm) against the golden reference model.

4 Synthesis Results

- **Technology node:** TSMC 180 nm
- **Area:** 69,900.971 μm^2 (2,823 mapped cells)
- **Max frequency:** 131.072 kHz (clock period 7,629,394.531 ps), the specified system clock; timing was met with a reported setup slack of +7,628,779 ps
- **Critical path:** `epsilon_R0_reg[23]/CK` $\rightarrow$ D, dominated by the register clock-to-Q delay of approximately 166 ps

The design was synthesized using Cadence Genus Synthesis Solution 21.10-p002_1, targeting the TSMC 180 nm `typical.lib` corner at the nominal 1.8 V/25°C operating point, with the top-level module `freq_diff_estimator` and the 131.072 kHz clock constraint applied uniformly across all four candidate RTL versions for a fair comparison. Synthesis proceeded through the

standard `syn_generic`, `syn_map`, and `syn_opt` stages, followed by generation of clock, timing, area, power, and gate reports and the post-synthesis Verilog netlist and SDF file.

Table 2: Synthesis results for the selected final RTL, Version C.

Parameter	Result
Technology node	TSMC 180 nm
Synthesis tool	Cadence Genus 21.10-p002_1
Library corner	typical.lib
Cell area	69,900.971 μm^2
Mapped cells	2,823
Sequential cells	25
Vectorless power	22.6098 μW
Clock period	7,629,394.531 ps
Equivalent frequency	131.072 kHz
Reported setup slack	+7,628,779 ps
Timing status	Met

4.1 Comparison Against Earlier Candidates

Table 3 compares Version C’s synthesis results against the earlier candidates evaluated in the same environment.

Table 3: Synthesis comparison of the evaluated RTL versions.

Version	Cell area (μm^2)	Cell count	Power (μW)
B	59,522.603	1,705	18.5601
A	80,332.562	2,359	34.4394
A2	98,168.719	2,368	47.9266
C	69,900.971	2,823	22.6098

Relative to Version A2, Version C reduces cell area by 28.80% and vectorless power by 52.82%. Relative to the original Version A, Version C reduces cell area by 12.99%.

4.2 Area Bottleneck: Dominant Arithmetic Cells

The Version A2 gate report showed that the design was dominated by arithmetic compressor and full-adder cells, as summarized in Table 4.

Table 4: Dominant arithmetic cells in Version A2.

Cell type	Instance count	Cell area (μm^2)
CMPR42X1	463	52,364.190
ADDFX2	281	19,629.087
Combined	744	71,993.277

These two arithmetic cell types alone occupied approximately 73.3% of Version A2’s total cell area, confirming that the wide reciprocal constant multiplier was the primary target for RTL optimization. Increasing the coefficient resolution (Version A $\rightarrow$ A2) had improved functional

accuracy, but had also increased the constant multiplier width, the product width, the number of compressor and full-adder cells, and switching activity in the combinational datapath.

In Version C, the same two cell types are reduced to 187 and 141 instances respectively – a 59.6% and 49.8% reduction relative to Version A2 – confirming that the algebraic simplification to the exact 32/25 factor removed the dominant area contributor rather than merely shrinking it.

All four versions produced a positive reported setup slack of +7,628,779 ps at the 131.072 kHz clock constraint; no timing degradation that would prevent operation at the specified frequency was observed for any candidate. The final input/output delays and drive strengths depend on the surrounding system integration environment and were not available during Week 3, so this timing comparison is limited to the common internal clock constraint.

5 Gate-Level Simulation

- **Pre-syn vs post-syn matching:** Pass
- **Issues found:** None

Post-synthesis verification was performed using Cadence Xcelium and SimVision. The Version C RTL was replaced by the mapped standard-cell netlist generated by Genus, `freq_diff_estimator_vC_postsyn.v`, and timing information was back-annotated using the corresponding SDF file, `freq_diff_estimator_vC_postsyn.sdf`. The simulation was performed using the TSMC 180 nm standard-cell Verilog simulation models and the same testbench clock period of 7,629.394531 ns used for functional verification.

At RTL, 1,000,000 constrained-random vectors had already verified the arithmetic; repeating the complete regression at gate level was impractical because the simulator must evaluate thousands of standard-cell instances and schedule their propagation-delay events. The gate-level regression therefore retained all directed, boundary, saturation, reset, handshake, and cancellation-sensitive cases, together with a reduced 1,000-vector constrained-random subset, as summarized in Table 5.

Table 5: Functional regression depth used at each verification level.

Simulation level	Random vectors	Total vectors
RTL functional regression	1,000,000	1,000,097
Zero-delay gate-level simulation	1,000	1,097
SDF-annotated gate-level simulation	1,000	1,097

Two gate-level simulations were run, each checking 1,097 vectors (all 97 directed/boundary/saturation/cancellation vectors plus 1,000 constrained-random vectors):

- **Zero-delay gate-level simulation:** completed with 0 recorded failures; the mapped netlist matched the golden model for all 1,097 checked vectors, with reset behavior, output saturation, and the `busy/done` handshake unchanged after technology mapping.
- **SDF-annotated timing simulation:** completed with 0 functional failures and 0 numerical difference from the golden result for all checked vectors, using maximum-delay values from the SDF file. No SDF file-access or instance-annotation error was observed.

A waveform transition corresponding to the critical path reported by static timing analysis was identified in SimVision, where `epsilon_R0` transitioned from 24'h800000 to 24'h7FFFFFFF

(sign bit `epsilon_R0`[23] falling). The two waveform cursors measured $t_{clk} = 240,325,911$ ps and $t_Q = 240,326,077$ ps, giving a measured clock-to-Q delay of

$$t_Q - t_{clk} = 240,326,077 - 240,325,911 = 166 \text{ ps},$$

which agrees exactly with the 166 ps falling clock-to-Q delay reported by Genus static timing analysis for `epsilon_R0_reg`[23], and corresponds to approximately 0.0022% of the 7,629,394.531 ps clock period.

The zero-delay and SDF-annotated simulations both passed, so the selected Version C implementation satisfies the following post-synthesis criteria:

- The synthesized netlist preserves the RTL arithmetic behavior.
- Output saturation remains correct at both signed Q1.23 limits.
- Reset, `update_en`, `busy`, and `done` behavior is preserved.
- The standard-cell propagation delays do not cause a functional failure at the specified clock frequency.
- The measured 166 ps output-register delay agrees with the static timing result.

6 Integration Notes

- **Interfaces to other modules:**

- `T_est` (signed 16-bit Q8.8) – from the Dual-RO Temperature Engine.
- `TC1`, `TC2` (signed 16-bit) – OTP-stored per-die calibration coefficients.
- `epsilon_aging` (signed 24-bit Q1.23) – from the rRO Aging Engine.
- `epsilon_R0` (signed 24-bit Q1.23) – output consumed by the downstream Sigma-Delta Modulator, which uses it to correct the fractional-N divider and the generated clock frequency.
- `update_en`, `busy`, `done` – the single-cycle update handshake.

- **Assumptions made:**

- `TC1` and `TC2` are loaded during power-on and remain static during normal operation, so they are treated as ordinary combinational inputs rather than internally latched registers.
- ϵ_{aging} is supplied already in Q1.23 format, consistent with the official golden reference model used by the course testbench, which performs no additional scaling on ϵ_{aging} before summation with the temperature terms.
- ϵ_{RO} is interpreted by the downstream Sigma-Delta Modulator as a signed 24-bit Q1.23 value.

6.1 DRS Requirements Preserved During Optimization

All four RTL candidates evaluated in Week 3 retained the same external interface and control behavior defined by the DRS; the optimization across versions was limited strictly to the arithmetic implementation of the quadratic temperature term. This is summarized in Table 6.

Table 6: DRS requirements preserved during RTL optimization.

Requirement	Implementation
System clock	131.072 kHz clock retained in all synthesis runs.
Reset	Active-low asynchronous reset retained.
Temperature input	Signed 16-bit Q8.8 <code>T_est</code> input retained.
Valid temperature range	Directed and constrained-random tests use -40°C to $+85^{\circ}\text{C}$.
Temperature coefficients	Signed 16-bit TC1 and TC2 inputs retained.
Aging input	Signed 24-bit Q1.23 <code>epsilon_aging</code> input retained.
Linear term	Exact scaling $TC1 \times dT \times 2^{13}$ retained.
Quadratic term	Mathematical function retained, while the constant-scaling architecture was modified (Versions B/A/A2/C).
Output format	Signed 24-bit Q1.23 <code>epsilon_R0</code> retained.
Saturation	Output remains clamped to $[-2^{23}, 2^{23} - 1]$.
Handshake	<code>busy</code> and <code>done</code> assert during an enabled update and clear afterward.

7 Issues and Resolutions

Table 7: Issues Identified and Their Resolutions

Issue	Resolution	Status
Version B used a low-resolution quadratic-scale constant ($328/2^8$), which failed to maintain the required 20 ppm tolerance for some test vectors.	Version B was retained as an area-accuracy trade-off experiment, but was not selected for the final implementation.	Closed
Version A used the reciprocal constant ($171,798,692/2^{27}$). Although it passed the smaller Week 2 regression, 8 cancellation-sensitive vectors exceeded the 168-LSB tolerance when the regression was expanded to 1,000,000 constrained-random vectors, with a maximum difference of 2,569 LSBs.	Version A was replaced by Version A2, which uses the higher-resolution reciprocal constant ($43,980,465,111/2^{35}$). Version A2 passed the complete regression with a maximum difference of only 3 LSBs.	Closed
Version A2 resolved the cancellation-related accuracy failures, but its wide reciprocal-constant multiplier became the dominant contributor to synthesized area and power, occupying $98,168.719 \mu\text{m}^2$ and $47.9266 \mu\text{W}$, respectively. Two arithmetic cell types alone accounted for approximately 73.3% of the total synthesized area.	The quadratic-term fixed-point expression was algebraically simplified to the exact factor $32/25$, leading to Version C and eliminating the need for the wide approximate reciprocal multiplier.	Closed

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Issue	Resolution	Status
It was necessary to verify that the algebraic simplification introduced in Version C did not cause any numerical deviation from the validated Version A2 implementation.	Version C was re-verified against the complete 1,000,097-vector regression and produced zero maximum difference from the golden model: 0 LSB and 0 ppm.	Closed
A discrepancy was identified between the DRS prose description and the official golden model concerning whether the temperature-dependent terms require an additional ppm-to-fraction conversion before being summed with ϵ_{aging} .	The final RTL implementation follows the convention specified by the official Section 7.2 golden model, without applying an additional conversion. This interpretation was confirmed by the course testbench.	Closed
Executing the complete 1,000,000-vector RTL regression at gate level was impractical because of the large simulation overhead associated with per-instance propagation-delay event scheduling.	The gate-level regression was reduced to 1,097 vectors, comprising 97 directed, boundary, saturation, and cancellation cases together with 1,000 constrained-random vectors. This reduced regression preserved the relevant functional corner cases. Both zero-delay and SDF-annotated gate-level simulations completed successfully with zero recorded failures.	Closed

8 Recommendations

- **For system integration:**

- Confirm with the system team that ϵ_{aging} is interpreted consistently as a pre-scaled Q1.23 fraction at the module interface, since this same ambiguity exists in the DRS text itself and could plausibly have been resolved differently by other groups.
- Request an official DRS erratum clarifying the inconsistency between the prose description and the Section 7.2 golden model regarding the ppm-to-fraction conversion of the temperature terms.
- Complete cross-group integration and top-level simulation with the modules supplying `T_est`, `TC1`, `TC2`, and `epsilon_aging`, and with the downstream module receiving `epsilon_R0`, as planned for Week 4.

- **For future improvements:**

- If a future revision targets a substantially higher operating frequency such that the true-division implementation of the quadratic term no longer fits within the available timing margin, a reciprocal-multiplier scheme could be reconsidered, provided it is re-verified against the full cancellation-case vector set that previously exposed the failure of Versions A and B.
- Organize the final RTL, testbench, constraints, synthesis reports, post-synthesis files, and simulation logs into the required Week 4 submission package.

Table 8: Week 4 final-submission package.

Item		Planned content
Final RTL code		<code>freq_diff_estimator_vC.v</code>
Final testbench		<code>tb_freq_diff_estimator.v</code>
Clock constraint		<code>fde_clock_const.tcl</code>
Synthesis script		Final reusable Cadence Genus synthesis script
Synthesis reports		Area, timing, power, gates, clocks, and design-check reports
Post-synthesis files		Version C mapped netlist and SDF file
Simulation logs		RTL, zero-delay gate-level, and SDF-annotated logs
Final report		Updated version of this report including Week 4 integration results
Code-package	docu-	README containing file descriptions and simulation/synthesis commands
mentation		

9 Deliverables Checklist

- ☒ RTL code (`freq_diff_estimator_vC.v`)
- ☒ Testbench code (`tb_freq_diff_estimator.v`)
- ☒ Synthesis script (`.tcl`)
- ☒ Synthesis report (area, timing) – Version C
- ☒ Gate-sim log – Version C (zero-delay and SDF-annotated)

All deliverables were completed during Week 3 for the selected Version C implementation. This is consistent with the Week 3 and 4 report’s own internal tracking, which listed eight technical deliverables (final RTL candidate, self-checking testbench, RTL regression, synthesis reports, post-synthesis outputs, gate-level simulation, simulation logs, and the Week 3 and 4 progress report) as complete. described in Table 8 and Section 8.

Signed

A: _____

B: _____

C: _____

D: _____